The Exact Extremal Bound in Erdős Problem 848

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Abstract

For $N \geq 1$, let $A \subseteq [1, N]$ satisfy the condition that $ab + 1$ is nonsquarefree for every $a, b \in A$, with $a = b$ allowed. We prove the sharp bound

$$|A| \leq \left\lfloor \frac{N+18}{25} \right\rfloor,$$

with equality attained by the elements of the progression $7 \pmod{25}$ in $[1, N]$. Earlier work established the same formula only above a large threshold. Our proof converts the extremal problem, by an exact Hall equivalence, into a completion inequality relative to the progression $7 \pmod{25}$. A prefix-compatible colouring controls all $N \leq 5{\cdot}10^6$ simultaneously. Beyond this point, square-divisor estimates first eliminate defects supported on the competing progression $18 \pmod{25}$ and then force every remaining defect to contain a large residual set. A valuation and residue decomposition supplies bounded pivot configurations; finite-prime counts and spacing between solutions of a transformed quadratic equation rule out each configuration. The resulting interval estimates are uniform, and a final monotone argument controls the unbounded tail. All finite inequalities and witnesses are exact, and the complete theorem is formally verified in Lean 4.

1 Introduction

Write

$$C_N = \{n \in [1, N] : n \equiv 7 \pmod{25}\}.$$

The quantifiers in the problem include the diagonal case $a = b$. Moreover,

$$|C_N| = \left\lfloor \frac{N+18}{25} \right\rfloor. \tag{1}$$

Erdős and Sárközy asked whether this construction is extremal [2]. Sawhney proved the assertion for all sufficiently large N by a density sieve and structural reduction [5]. Sothanaphan made the asymptotic inputs explicit and obtained the threshold $N \geq 2.64 \cdot 10^{17}$ [6]. We prove the exact statement

$$|A| \leq |C_N| \tag{2}$$

for every positive integer N.

Theorem 1.1 (Exact solution of Erdős Problem 848)**.** *For every integer $N \geq 1$, the maximum cardinality of a set $A \subseteq [1, N]$ for which $ab + 1$ is nonsquarefree for every $a, b \in A$ is*

$$|\{1 \leq n \leq N : n \equiv 7 \pmod{25}\}| = \left\lfloor \frac{N + 18}{25} \right\rfloor .$$

Already at $N = 18$ the two residue classes that organize the proof are visible. Both singleton sets $\{7\}$ and $\{18\}$ are admissible because $7^2 + 1$ and $18^2 + 1$ are divisible by 25, whereas $7 \cdot 18 + 1 = 127$ is squarefree. The theorem says that no admissible pair exists. Thus the exact problem is not merely to count one progression; it is to control how alternative residue classes can interact with it.

Work	Proven range and conclusion	Structural information	Role in the exact theorem
Sawhney (2025)	The class 7 mod 25 is extremal for every $N \geq N_0$, for some unspecified N_0; a stability statement identifies the two candidate extremal classes.	The stability argument identifies the two candidate extremal classes.	Provides the asymptotic structure on which the exact Hall analysis is built.
Sothanaphan (2026)	The same extremal bound holds explicitly for $N \geq 2.64 \cdot 10^{17}$.	The threshold is explicit, but the exact theorem below it remains open.	Provides quantitative squarefree-progression and large-divisor estimates.
This paper	The exact extremal value is proved for every $N \geq 1$.	The Hall reformulation identifies the obstruction at every scale.	Prefix-compatible colouring, valuation–residue pivots, transformed-root spacing, and a monotone unbounded-tail estimate.

Table 1: Prior results and the all-N theorem.

The proof has four layers. First, an exact Hall equivalence replaces the extremal-set question by one inequality for a completion $H_N(B)$. Second, a single interval-valued colouring simultaneously controls the entire initial range. Third, beyond that range, matching and degree estimates force a hypothetical Hall defect into a large residual set. Its 2-adic classes, residues modulo 9, and occasional fibres modulo 7^2 or 11^2 produce a bounded family of pivot configurations. Finally, quadratic-residue restrictions on the complementary quotient in $bx + 1 = p^2 m$, together with spacing between transformed roots, give uniform bounds on each structural interval. The endpoint comparisons are exact rational inequalities, and monotonicity propagates the final one to all larger N.

The squarefree-progression estimates are combined with exact residue and valuation information tailored to the Hall graph. The finite data consist of exhaustive residue and interval certificates governed by general soundness theorems.

Proposition 1.2 (Sharpness)**.** *The set C_N is admissible: if $a, b \in C_N$, then $ab + 1$ is not squarefree.*

Proof. Since $a \equiv b \equiv 7 \pmod{25}$,

$$ab + 1 \equiv 7^2 + 1 = 50 \equiv 0 \pmod{25}.$$

Thus $5^2 \mid ab + 1$. □

2 The exact Hall cut

Throughout, N is a positive integer and $[1, N] = \{1, \ldots, N\}$. An integer is *nonsquarefree* if it is divisible by p^2 for some prime p. Join $b \in [1, N] \setminus C_N$ to $a \in C_N$ when $ab + 1$ is squarefree. For $B \subseteq [1, N] \setminus C_N$, put

$$\Gamma_N(B) = \{a \in C_N : ab + 1 \text{ is squarefree for some } b \in B\}$$

and

$$T_N(B) = C_N \setminus \Gamma_N(B).$$

Thus $a \in T_N(B)$ exactly when $ab + 1$ is nonsquarefree for every $b \in B$. Call B a *compatible outside set* if

$$B \subseteq [1, N] \setminus C_N \quad \text{and} \quad bb' + 1 \text{ is nonsquarefree for every } b, b' \in B,$$

where $b = b'$ is allowed.

The usual Hall condition for the squarefree-edge graph is $|\Gamma_N(B)| \geq |B|$. Since $|\Gamma_N(B)| = |C_N| - |T_N(B)|$, it is equivalent to

$$|B| + |T_N(B)| \leq |C_N|. \tag{3}$$

It suffices to test compatible outside sets, since these are precisely the possible outside portions of an admissible set.

Theorem 2.1 (Hall reduction)**.** *If (3) holds for every compatible outside set B, then (2) holds.*

Proof. Let A be admissible and put $B = A \setminus C_N$. Then $A \cap C_N \subseteq T_N(B)$, whence

$$|A| = |B| + |A \cap C_N| \leq |B| + |T_N(B)| \leq |C_N|.$$

□

Theorem 2.2 (Hall equivalence)**.** *The Hall inequality (3) is equivalent to the exact upper bound (2).*

Proof. The forward implication is Theorem 2.1. Conversely, assume (2) and let B satisfy the hypotheses of (3). Put $A = B \cup T_N(B)$. Products internal to B are nonsquarefree by hypothesis; products internal to $T_N(B)$ are divisible by 25; and every cross-product is nonsquarefree by the definition of $T_N(B)$. Hence A is admissible. The two parts are disjoint, so

$$|B| + |T_N(B)| = |A| \leq |C_N|.$$

□

For later normalization define the Hall completion

$$H_N(B) = B \cup T_N(B).$$

The union is disjoint, so $|H_N(B)| = |B| + |T_N(B)|$. The exact lower bound

$$|C_N| \geq \frac{N}{25} - \frac{7}{25} \tag{4}$$

shows that

$$\frac{|H_N(B)|}{N} \leq \frac{1}{25} - \frac{7}{25N}$$

implies (3). All rational comparisons below are made against this target. We call a compatible outside set with $|C_N| < |B| + |T_N(B)|$ a *strict Hall defect*.

3 A simultaneous prefix colouring

Theorem 3.1 (Exact prefix colouring)**.** *For every $1 \le N \le 5 \cdot 10^6$, let*

$$\mathcal{D}_N = \{x \in [1, N] : x^2 + 1 \text{ is nonsquarefree}\},$$

and join distinct $x, y \in \mathcal{D}_N$ when $xy + 1$ is nonsquarefree. This graph has a proper colouring whose colour set is indexed by C_N. Consequently (2) and (3) hold throughout this interval.

Proof. The diagonal hypothesis shows that every member of an admissible set belongs to $\mathcal{D}_N$, and the off-diagonal hypothesis makes the admissible set a clique in this graph. Thus a proper colouring by at most $|C_N|$ colours implies the cardinality bound. Here "indexed by C_N" means a map into C_N; some colours may be unused. It is enough to specify the colouring by interval data, because the assignment changes only at finitely many prefix endpoints.

Put $L = 5 \cdot 10^6$. At a prefix write

$$U = \{u \le N : u \equiv 7 \pmod{25}\}, \qquad V = \{v \le N : v \equiv 18 \pmod{25}\}, \qquad E = \mathcal{D}_N \setminus (U \cup V).$$

For each $u \equiv 7 \pmod{25}$ and each $N \in [u, L]$, a state has three possible occupants: the anchor u, at most one element of V, and at most one element of E. Empty non-anchor positions are recorded by zero. The state is required to satisfy that every two positive occupants in different positions have squarefree product plus one. Consequently no edge of the nonsquarefree graph can have both endpoints in one state. The state is constant on a half-open interval of values of N, and the successive intervals form a contiguous partition of $[u, L + 1)$.

The reverse data for a candidate x form another contiguous interval partition, beginning at x, the first prefix at which x is present. On every interval they point to an anchor state whose occupant is exactly x. Candidates in U point to their own anchor; candidates in V are indexed by their position in the progression $18 \bmod 25$. For $x \in E$, diagonal completeness is proved separately: if $x^2 + 1$ is nonsquarefree, then some odd prime $p \equiv 1 \pmod 4$ satisfies $p^2 \mid x^2 + 1$, and a tabulated root, verified directly modulo p^2, of $z^2 \equiv -1 \pmod{p^2}$ places x in the indexed diagonal set.

Fixing N and following these reverse assignments defines a map $c_N : \mathcal{D}_N \to C_N$. The interval-cover conditions show that c_N is defined for every diagonal candidate. If $x \ne y$ and $c_N(x) = c_N(y)$, then they occupy different positions of the same anchor state, so the local squarefreeness proof gives $xy+1$ squarefree. Hence c_N is a proper colouring with colour set C_N, exactly as required. $\square$

4 Uniform Hall reductions beyond the prefix

Let

$$D_N = \{n \in [1, N] : n \equiv 18 \pmod{25}\}.$$

For a Hall completion $H_N(B)$ put

$$R_N(B) = H_N(B) \setminus (C_N \cup D_N).$$

We first eliminate defects supported only on D_N, and then obtain a lower bound for the residual set whenever it is nonempty.

4.1 Pure opposite-base matching

For each parity, split the source D_N into that parity and the target C_N into the opposite parity. The target block has at least the size of the source block: every $x \in D_N$ satisfies $x \geq 18$, and $x \mapsto x-11$ lies in $[1, N]$, changes the residue 18 to 7 modulo 25, and reverses parity. For opposite parities, $xy+1$ is odd, so the square prime 2 is inactive; moreover $xy+1 \equiv 18\cdot 7+1 \equiv 2 \pmod{25}$, so 5 is inactive.

For a fixed pivot, nonsquarefree values are divided into three disjoint prime ranges:

$$p \leq 47, \qquad 47 < p \leq N/26, \qquad p > N/26.$$

The following three lemmas bound the corresponding exceptional sets.

Lemma 4.1 (Small-prime contribution). *Let $N \geq 5 \cdot 10^6$, let $X \subseteq [1, N]$ lie in one residue class modulo 50, and let b be any pivot. Then*

$$\frac{\#\{x \in X : \exists p \in \{3, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\},\ p^2 \mid bx+1\}}{N} \leq \frac{S_0}{5 \cdot 10^6},$$

where

$$S_0 = 100001 \sum_p \frac{1}{p^2} + 13$$

and the sum is over the displayed thirteen primes.

Proof. For a fixed p, the conditions $x \bmod 50$ and $p^2 \mid bx+1$ select at most one class modulo $50p^2$, hence at most

$$\left\lfloor \frac{N+1}{50p^2} \right\rfloor + 1$$

points. Sum this bound over the thirteen primes. After division by N, the endpoint term is decreasing for $N \geq 5 \cdot 10^6$, so its maximum is the displayed value at $5 \cdot 10^6$. □

Lemma 4.2 (Medium-prime contribution). *Under the same hypotheses, the contribution of $47 < p \leq N/26$ is at most*

$$\frac{3887}{25 \cdot 10^6} + \frac{59}{10^4}.$$

Proof. The residue-class union bound gives the reciprocal-square contribution

$$\frac{1}{25} \sum_{47<p\leq N/26} \frac{1}{p^2}.$$

The reciprocal-square estimate used here is less than $3887/10^6$. The endpoint term contributes $\pi(N/26)/N$, and an explicit prime-count estimate bounds this by $59/10^4$ for $N \geq 5 \cdot 10^6$ [4, 1]. Adding the two terms proves the claim. □

Lemma 4.3 (High-prime contribution). *Assume in addition that $1 \leq b \leq N$ and that $bx+1$ is coprime to 10 for every $x \in X$. Then the contribution of primes $p > N/26$ is at most*

$$\frac{11264064/3125}{5 \cdot 10^6}.$$

Proof. Write $bx+1=p^2m$. Since $p>N/26$, we have $p\ge \lfloor N/26\rfloor+1$; while $b,x\le N$, one has

$$bx+1\le N^2+1<677\big(\lfloor N/26\rfloor+1\big)^2.$$

Hence $m\le 676$. After writing the fixed mod-50 progression $x=r+50t$, the equation becomes

$$p^2m=(br+1)+50bt. \tag{5}$$

Let

$$S(b)=\{q: q \text{ is prime},\ q\mid b,\ q\notin\{2,5\}\}.$$

Reducing (5) modulo $q\in S(b)$ gives $p^2m\equiv 1 \pmod q$. Thus $q\nmid m$ and m is a nonzero quadratic residue modulo q. Together with the parity and mod-5 condition, these congruences leave a finite set of possible $m\le 676$.

For fixed m, define the root classes to be the residues $s \pmod{50b}$ satisfying

$$s^2m\equiv br+1 \pmod{50b}.$$

Every possible prime p lies in one of these classes. If two solutions in one class have $p_1<p_2$, then $p_2-p_1\ge 50b$. Subtracting their two instances of (5) gives

$$50b(t_2-t_1)=m(p_2-p_1)(p_2+p_1),$$

and $p_1>N/26$ yields the spacing inequality

$$t_2-t_1\ge 2m(N/26).$$

Summing the resulting fibre bound over the admissible m and their root classes bounds the number of possible t. The finite calculation lists exactly those m and root classes satisfying the quadratic-residue conditions above, indexed by the prime-divisor set $S(b)$. Since the elements of $S(b)$ are distinct prime divisors of b, $\prod_{q\in S(b)} q\le b\le N$; hence every pivot is covered by one of these cases. Applying the spacing inequality in each case gives the coefficient 11264064/3125. After division by N, the resulting bound is largest at $N=5\cdot 10^6$. □

Theorem 4.4 (Uniform pure opposite-base matching)**.** *For every $N\ge 5\cdot 10^6$, the squarefree-edge graph from D_N to C_N has a matching saturating D_N. Consequently every compatible $B\subseteq D_N$ satisfies* (3).

Proof. The three contributions satisfy the exact rational inequality

$$\frac{S_0}{5\cdot 10^6}+\left(\frac{3887}{25\cdot 10^6}+\frac{59}{10^4}\right)+\frac{11264064/3125}{5\cdot 10^6}\le \frac{1}{100}-\frac{1}{5\cdot 10^6}. \tag{6}$$

Lemmas 4.1–4.3 therefore show, in each parity block and after interchanging source and target, that every vertex has squarefree degree greater than half of the opposite block. The elementary dense bipartite matching lemma now applies. Indeed, if a source subset has at most half the target size, the neighbourhood of any one of its vertices is already larger than the subset. If it has more than half the target size and missed a target vertex, all neighbours of that target would lie in a source complement of size less than half the source block, contradicting the reverse degree bound. Thus Hall's condition holds. The two parity matchings combine because their target parities are disjoint. Finally, a matching from $B\subseteq D_N$ into $C_N\setminus T_N(B)$ gives $|B|\le |C_N|-|T_N(B)|$. □

4.2 A quantitative mixed residual

Proposition 4.5 (Uniform mixed degree)**.** *Let $N \ge 5 \cdot 10^6$, let B be a compatible outside set, and let $b \in R_N(B)$. Then b has more than*

$$\frac{N}{50} + \frac{N}{525}$$

squarefree neighbours in each of C_N and D_N.

Proof. Since $T_N(B) \subseteq C_N$, removing $C_N \cup D_N$ from $H_N(B)$ leaves exactly $B \setminus D_N$. Thus $b \in B$ and $b \not\equiv 7, 18 \pmod{25}$. For $x \equiv 7$ or $18 \pmod{25}$, the congruence $25 \mid bx + 1$ would force, respectively, $b \equiv 7$ or $18 \pmod{25}$. Hence the square 5^2 cannot divide either base form associated with a residual pivot. Split a nonsquarefree form $bx + 1$ into the three ranges

$$p \in \{2, 3, 7\}, \qquad 7 < p \le N/55, \qquad p > N/55.$$

The constants

$$\tau_7 = \frac{263529083909042886517376461184337967}{8573456796637692379906289787841000000}, \qquad \rho_7 = \frac{221926420176}{12755647965025}.$$

have the following meanings: τ_7 bounds the reciprocal-square sum for primes larger than 7, and ρ_7 bounds the number of admissible roots of the transformed equations. For $p \in \{2, 3, 7\}$, simultaneous reduction modulo $25p^2$ gives the exact normalized upper bound $17/1225$. For $7 < p \le N/55$, the union bound over the unique relevant residue class gives $\tau_7/25$.

For $p > N/55$, write $x = r + 25t$ with $r \in \{7, 18\}$ and $bx + 1 = p^2 m$. The same comparison with $N^2 + 1$ now gives $m \le 55^2 = 3025$. There are two transformed equations. If $5 \nmid br + 1$, then $5 \nmid m$ and

$$p^2 m = (br + 1) + 25bt.$$

If $5 \mid br + 1$, then $p \ne 5$ forces $5 \mid m$; writing $m = 5m'$ gives $m' \le 605$ and, after division by 5,

$$p^2 m' = \frac{br + 1}{5} + 5bt.$$

We call these the normal and mod-5-twisted equations, respectively. The nonzero quadratic-residue conditions modulo the prime divisors of b, treated separately in the three even 2-adic classes and the two odd classes, give the common upper bound $\rho_7/6$. The base progression contains at least $N/25 - 1$ points. The exact margin is

$$\frac{17}{1225} + \frac{\tau_7}{25} + \frac{\rho_7}{6} + \frac{1 + 19601/49}{5 \cdot 10^6} < \frac{1}{50} - \frac{1}{525}.$$

The endpoint correction decreases with N, so subtracting all bad points from the base progression gives the stated degree in both directions. □

Corollary 4.6 (Uniform mixed residual)**.** *If $N \ge 5 \cdot 10^6$ and a compatible outside set B violates (3), then*

$$\frac{2N}{525} - 1 < |R_N(B)|. \tag{7}$$

In particular $|R_N(B)| > 19046$, and one of the five valuation parts used below has more than 3809 elements.

Proof. Theorem 4.4 implies that $R_N(B)$ is nonempty; choose $b \in R_N(B)$. Apply Proposition 4.5 to both base classes. Let d_7 and d_{18} be the resulting squarefree degrees. The C_N-neighbours are disjoint from $T_N(B)$, and the D_N-neighbours are disjoint from $B \cap D_N$, because products internal to B are nonsquarefree. Combining these two facts with the strict defect and the partition $B = (B \setminus D_N) \dot\cup (B \cap D_N)$ gives the exact degree-subtraction inequality

$$d_7 + d_{18} < |B \setminus D_N| + |D_N|.$$

Here $B \setminus D_N = R_N(B)$ and $|D_N| \leq N/25 + 1$. Since both degrees are strictly larger than $N/50 + N/525$, we obtain

$$\frac{2N}{525} - 1 < |R_N(B)|.$$

At $N = 5 \cdot 10^6$ the left side is greater than 19046, and it increases with N. The five valuation parts are disjoint and cover $R_N(B)$. If each had size at most 3809, their union would have size at most $5 \cdot 3809 = 19045$, a contradiction. □

5 Valuation classes and pivot estimates

The five valuation classes record the 2-adic information needed by the congruence counts. They are

$$E_1,\ E_2,\ E_3,\ O_1,\ O_3,$$

where, respectively,

$$x \equiv 2 \pmod 4, \quad x \equiv 4 \pmod 8, \quad x \equiv 0 \pmod 8, \quad x \equiv 1 \pmod 4, \quad x \equiv 3 \pmod 4.$$

Thus E_1, E_2, E_3 mean $v_2(x) = 1$, $v_2(x) = 2$, and $v_2(x) \geq 3$, respectively, while O_1, O_3 are the two odd residues modulo 4. These five conditions are disjoint and exhaustive. Within a class we next split by the nine residues modulo 9. This records exactly when 3^2 can divide several pivot forms simultaneously. When a finer separation is required, we use fibres modulo 7^2 or 11^2 for the same reason: they control simultaneous divisibility by the next small prime squares.

Lemma 5.1 (Valuation partition)**.** *The five valuation parts are pairwise disjoint and their union is $R_N(B)$. Consequently their cardinalities sum to $|R_N(B)|$.*

Proof. Every even integer satisfies exactly one of $v_2(x) = 1$, $v_2(x) = 2$, or $v_2(x) \geq 3$, and every odd integer is congruent to exactly one of 1 and 3 modulo 4. These alternatives are pairwise disjoint and exhaustive, so the cardinalities add. □

The next lemma treats the odd valuation classes in the five-to-ten-million range. It distinguishes a set spread across sufficiently many fibres from a concentration that admits a direct cardinality bound.

Lemma 5.2 (Nine-cell capacity dichotomy)**.** *Fix one odd valuation class after subsets already covered by earlier bounds have been removed, and let $\mathcal{R}$ be the remaining set. Join a residue label $c \pmod 9$ to a residue $r \pmod{49}$ when some $x \in \mathcal{R}$ satisfies $x \equiv c \pmod 9$ and $x \equiv r \pmod{49}$. For a set Y of mod-9 labels, let $\Gamma_{49}(Y)$ be its neighbourhood in the mod-49 residues. Then either eight distinct mod-9 labels can be assigned occupied mod-49 fibres, with no fibre used more than twice, or there is a set Y for which*

$$2\,|\Gamma_{49}(Y)| + 1 < |Y|.$$

Proof. Apply Hall's theorem to the bipartite graph whose left side is the nine mod-9 labels and whose right side consists of two copies of every mod-49 residue together with one additional vertex. A saturating matching uses the additional vertex at most once. Every other matched edge has, by definition, an occupant in $\mathcal{R}$; choosing one produces eight pivots in distinct mod-9 cells, with each mod-49 residue used at most twice. If no saturating matching exists, Hall supplies Y with the displayed capacity defect. $\square$

The counting identity used after the pivots are selected begins with the disjoint partition

$$H_N(B) = R_N(B) \,\dot{\cup}\, \big(H_N(B) \cap (C_N \cup D_N)\big).$$

Every element of $H_N(B)$ has nonsquarefree product plus one with every other element of $H_N(B)$: this follows from compatibility inside B, the definition of $T_N(B)$ for cross-products, and divisibility by 25 inside $T_N(B) \subseteq C_N$. Fix three pivots $P = \{p_1, p_2, p_3\}$ in the completion and a prime cutoff z. For a base point x, let $k(x)$ be the number of pivots for which $xp_i + 1$ has a square-prime divisor at most z, and let $\ell(x) = 3 - k(x)$ count the remaining large-prime events. Pointwise,

$$1 \le \frac{1}{2}\mathbf{1}_{k(x)=3} + \frac{1}{2}\mathbf{1}_{k(x)\ge 2} + \frac{1}{2}\ell(x).$$

Indeed, the three possible ranges $k = 3$, $k = 2$, and $k \le 1$ make the right-hand side at least one. Summing gives the exact three-pivot inequality

$$\begin{aligned} |H_N(B)| \le |R_N(B)| &+ \frac{1}{2}\#\{x : \text{all three finite-prime events occur}\} \\ &+ \frac{1}{2}\#\{x : \text{at least two finite-prime events occur}\} \\ &+ \frac{1}{2}\sum_{p\in P}\#\{x : \exists q > z \text{ prime with } q^2 \mid px + 1\}. \end{aligned} \tag{8}$$

The points x in the three counted sets are restricted to the base part $H_N(B) \cap (C_N \cup D_N)$. The smaller ranges also use four- and six-pivot versions of this same double-counting principle; from twenty million onward, all ten cases use exactly (8).

For a case β, let $P_\beta \subseteq R_N(B)$ be its selected pivots. After division by N, the residual bound and the three terms in (8) give an inequality of the form

$$\frac{|H_N(B)|}{N} \le D_\beta + F_\beta + Q_\beta < \frac{1}{25} - \frac{7}{25L} \le \frac{1}{25} - \frac{7}{25N}. \tag{9}$$

Here L is the lower endpoint of the interval under consideration. The term D_β bounds $|R_N(B)|/N$. It is a diagonal bound because every $x \in R_N(B) \subseteq H_N(B)$ satisfies that $x^2 + 1$ is nonsquarefree; the valuation and mod-9 information of the selected case sharpens this count. Any valuation classes already known to be small contribute their explicit cardinality bounds to D_β. The term F_β comprises the first two finite-prime counts in (8), and Q_β is the large-prime sum, bounded by reciprocal-square estimates and the transformed-root spacing argument. The middle inequality is an exact rational comparison, while the last follows because $1/25 - 7/(25N)$ increases with N. Equation (9) therefore contradicts a strict Hall defect.

Theorem 5.3 (The five-to-ten-million range)**.** *The Hall inequality holds for*

$$5 \cdot 10^6 \le N < 10^7.$$

Proof. Assume a strict defect. Corollary 4.6 gives $|R_N(B)| > 19046$. This exceeds the total number removed in any of the case reductions below; the largest such total is 128. We consider the five valuation classes in turn.

First examine E_1. Its mod-9 cells and mod-49 or mod-121 fibres either give a three- or four-pivot configuration, or imply $|E_1| \le 10$. Under the latter assumption, the nine-cell dichotomy for E_2 gives either two occupied cells, one sufficiently occupied cell, or $|E_2| \le 10$. If both classes satisfy these bounds, 19 points in E_3 give the next three-pivot configuration; otherwise $|E_3| \le 18$. Thus all three even classes together contribute at most

$$10 + 10 + 18 = 38$$

points in the only case that reaches the odd classes.

If one odd class has at most 45 points, then at most $38 + 45 = 83$ points lie outside the other odd class. The residual lower bound leaves that other class nonempty after its mod-9 cells of size at most 5 have been removed: more than 128 residual points and at most 83 outside the class leave more than 45 inside it, whereas the nine sparse cells contain at most $9 \cdot 5 = 45$ points. Hence at least one mod-9 cell contains at least six points. The number of such cells is between one and nine. The corresponding pivot estimate is proved separately for each possible number, with a common estimate for six through nine cells. Otherwise both odd classes contain at least 46 points. In each class, remove every mod-9 cell of size at most 5; this removes at most $9 \cdot 5 = 45$ points per class. Together with the even classes, at most

$$38 + 45 + 45 = 128$$

points have been removed, so a mod-9 cell of size at least six remains. Either the remaining dense cells contain disjoint triples, giving the six-pivot case, or their union has at most 11 cells and yields one of the four-pivot cases. These alternatives exhaust the possibilities by the nine-cell capacity dichotomy.

In every alternative the chosen pivots lie in $H_N(B)$ and the corresponding finite-prime, transformed-root, and diagonal estimates give (9) with $L = 5 \cdot 10^6$. The strict rational inequality contradicts the assumed defect. ∎

Theorem 5.4 (The ten-to-twenty-million range)**.** *The Hall inequality holds for*

$$10^7 \le N < 2 \cdot 10^7.$$

Proof. Again $|R_N(B)| > 19046$. Put

$$c_N = \left\lfloor \frac{N-1}{1000001} \right\rfloor + 1;$$

for $N < 2 \cdot 10^7$ one has $c_N \le 20$. More than c_N points in one mod-9 cell contain a pair at distance at most 10^6. In E_1 and then E_2, the case analysis first asks whether two distinct cells have more than c_N points, producing two close pairs and hence four pivots. If not, it asks whether one cell has more than c_N points, producing a three-pivot configuration while all other cells have size at most c_N. If neither happens, the entire valuation class has size at most $9c_N$. The same one-cell test is then applied to E_3.

If all three even classes satisfy these bounds, their union has size at most $27c_N$. The valuation partition and residual lower bound ensure that at least 91 residual pivots remain in the two odd classes. If one odd class has at most 45 elements, the other has at least 46; if neither does, both have at least 46. The corresponding one-class or two-class argument gives a three-pivot configuration. Thus the two-dense-cell alternatives give four pivots, while the one-cell and odd alternatives give three, and the cases are exhaustive.

For each of these alternatives, the selected three or four pivots give the same decomposition as in (9). Direct residue counts give F_β, the prime-square fibres give Q_β, and the bounds for the remaining cells are included in D_β. Substitution puts the total strictly below $1/25 - 7/(25N)$, so a strict defect is impossible. $\square$

5.1 The global ten-case decomposition

The later ranges use the following exhaustive case decomposition. Put

$$g_N = \left\lfloor \frac{N-1}{20001} \right\rfloor + 1, \qquad \delta = \frac{1}{20001} + \frac{1}{2 \cdot 10^7}.$$

The number g_N is an exact cardinality cap, and $g_N/N \le \delta$ for $N \ge 2 \cdot 10^7$. A class containing more than g_N points has two points at distance at most 20000 and, since $g_N \ge 2$, a third distinct point. Call the resulting triple *generic* when its three members are not all congruent modulo 9. Otherwise, either the whole valuation class lies in that common mod-9 cell, or a point outside the cell can replace the third member and make the triple generic. Hence, if no generic triple exists, the entire valuation class lies in one residue class modulo 9.

Lemma 5.5 (Ten-case decomposition)**.** *Let $N \ge 2 \cdot 10^7$ and suppose that B is a compatible outside set with a strict Hall defect. Then one of the ten cases in Table 2 applies.*

Proof. The uniform cutoff-19 degree theorem strengthens the mixed-residual estimate to $0.0045N < |R_N(B)|$. Inspect E_1, E_2, E_3 in order. If the current class has more than g_N members, the preceding close-triple and mod-9 dichotomy gives one of its two cases. Otherwise the class has cardinality at most g_N. If all three even classes satisfy this bound, the remaining residual elements lie in the two odd classes. When both odd classes are nonempty, the pigeonhole principle on intervals of length 460 gives a pair at distance at most 459; when only one odd class remains, intervals of length 230 give a pair at distance at most 229. Adding a third point and applying the same mod-9 dichotomy gives the final four cases. These alternatives exhaust all five valuation classes. $\square$

The eight symbols in the last column of Table 2 are the eight diagonal subsets that are counted separately: all residual points; one common E_1 cell; the low-2-adic classes; one common E_2 cell; the union of the two odd classes; that union with one common cell; one odd class; and one odd class with a common cell. The entry $j\delta$ accounts for at most j even classes satisfying the preceding size bounds.

Theorem 5.6 (The twenty-to-forty-million range)**.** *The Hall inequality holds for*

$$2 \cdot 10^7 \le N < 4 \cdot 10^7.$$

Proof. Lemma 5.5 gives one of the ten cases in Table 2. The six even cases use finite cutoff 23; the four odd cases use cutoff 19. For the pivots selected in case β, the decomposition above gives

$$\frac{|H_N(B)|}{N} \le T_\beta,$$

Case	Condition after the preceding size bounds hold	Set used to bound D_β
E_1 generic	close triple in E_1, not constant mod 9	all residual points
E_1 common	all of E_1 lies in one mod-9 cell	common E_1 cell
E_2 generic	$\lvert E_1\rvert \le g_N$; generic triple in E_2	all residual points
E_2 common	$\lvert E_1\rvert \le g_N$; all of E_2 lies in one cell	common E_2 cell $+\delta$
E_3 generic	$\lvert E_1\rvert, \lvert E_2\rvert \le g_N$; generic triple in E_3	$E_1 \cup E_2$, plus 2δ
E_3 common	same, with all of E_3 in one cell	$E_1 \cup E_2$, plus 2δ
two odd, generic	all even classes have size $\le g_N$; generic odd triple	$O_1 \cup O_3$, plus 3δ
two odd, common	same, with the relevant odd class in one cell	$O_1 \cup O_3$ and that cell, plus 3δ
one odd, generic	all even classes are small; the other odd class is empty	remaining odd class, plus 3δ
one odd, common	same, with the remaining odd class in one cell	that common cell, plus 3δ

Table 2: The ten exhaustive cases used for $N \ge 2 \cdot 10^7$.

where $T_\beta = D_\beta + F_\beta + Q_\beta$. To make the normalization and strict margin explicit, we display one complete comparison. Consider the case in which E_1 has at most $g_N = \lfloor (N-1)/20001 \rfloor + 1$ elements and E_2 contains a triple that is not contained in one residue class modulo 9. Put

$$\tau_{23} = \frac{640818027476480356298 63}{759666844402282624900 0000}.$$

This is the reciprocal-square upper bound for primes larger than 23. In this case the diagonal estimate for all residual points already includes E_1, so no separate δ term is needed. The three upper bounds in (9) are

$$D_\beta = \frac{25289550}{10^9}, \qquad F_\beta = \frac{8685}{10^6}, \qquad Q_\beta = \frac{3\tau_{23}}{25} + \frac{1}{2}\frac{10006474}{10^9}.$$

Consequently

$$T_\beta = \frac{30379114085046090894711 4523}{75966684440228262490000000 00},$$

$$\left(\frac{1}{25} - \frac{7}{25 \cdot 2 \cdot 10^7}\right) - T_\beta = \frac{754905570939246933179 91}{75966684440228262490000000 00} > 0.$$

The other nine cases differ only in the valuation classes already bounded, the diagonal subset from Table 2, and the finite- and large-prime bounds. Substituting their exact rational values proves, in every case,

$$T_\beta < \frac{1}{25} - \frac{7}{25 \cdot 2 \cdot 10^7} \le \frac{1}{25} - \frac{7}{25N}$$

and hence contradicts the assumed Hall defect. $\square$

6 Intermediate-scale Hall bounds

Theorem 6.1 (The forty-to-two-hundred-million range)**.** *The Hall inequality holds for*

$$4 \cdot 10^7 \le N < 2 \cdot 10^8.$$

Proof. The interval is divided into the six half-open blocks with endpoints

$$40,\ 50,\ 70,\ 80,\ 100,\ 150,\ 200 \quad \text{million}.$$

Lemma 5.5 gives one of the ten cases. On each block, D_β is the corresponding diagonal upper bound plus the exact cardinality bounds for preceding small classes. The finite-prime terms F_β, in the order of Table 2, are

$$10^{-6}(8685, 12616, 8685, 12616, 8685, 12616, 19420, 20878, 26643, 29459).$$

The term Q_β uses cutoff 23 in the six even cases and cutoff 19 in the four odd cases, together with the transformed-root bound for the current block. These are uniform inequalities for every N in the block: the residue counts and fibre-spacing theorem give the first inequality in (9) uniformly throughout the block. The final rational comparison is made at the lower endpoint L, after which the monotonicity of $1/25 - 7/(25N)$ gives the result throughout the block. Hence no strict defect exists. □

Theorem 6.2 (The two-hundred-million-to-two-billion range)**.** *The Hall inequality holds for*

$$2 \cdot 10^8 \leq N < 2 \cdot 10^9.$$

Proof. The four subintervals have endpoints

$$200,\ 300,\ 500,\ 1000,\ 2000 \quad \text{million}$$

and transformed-root splits $75, 85, 100, 125$, respectively. Thus the large-prime threshold is $p > N/s$ with s equal to the listed split, and the complementary quotient is at most s^2. The normal and mod-5-twisted cases use, respectively, the integers in $[1, s^2]$, with division by 5 in the twisted case, that satisfy the nonzero quadratic-residue conditions forced by the prime divisors of the pivot. The fibre-spacing argument following (5) converts their exact cardinalities into a root bound for each subinterval.

Lemma 5.5 gives one of the same ten cases. For a subinterval ρ and case β, define

$$T_{\rho,\beta} = D_{\rho,\beta} + F_\beta + Q_{\rho,\beta},$$

where $D_{\rho,\beta}$ is the bound for the relevant one of the eight diagonal subsets described before Table 2, F_β is the finite-prime vector displayed in the preceding proof, and $Q_{\rho,\beta}$ uses the subinterval's transformed-root upper bound

$$10^{-9}(9000000,\ 7700000,\ 6400000,\ 5000000).$$

The finite sets consist exactly of the complementary quotients satisfying the displayed congruence conditions. The equation $p^2 m = bx + 1$ and the prime divisors of b show that every possible quotient belongs to the appropriate set; the spacing lemma then gives its contribution to the cardinality bound. Combining this with the diagonal and finite-prime estimates proves $|H_N(B)|/N \leq T_{\rho,\beta}$, and exact rational arithmetic gives

$$T_{\rho,\beta} < \frac{1}{25} - \frac{7}{25L_\rho} \leq \frac{1}{25} - \frac{7}{25N}$$

for every subinterval and case. This contradicts a strict defect. □

7 The unbounded Hall regime

For the high tail the same ten mathematical cases remain valid. Only the uniform bounds for the relevant diagonal subset and for the admissible roots change with N, so the finite part is covered by consecutive intervals and the remaining part by one monotone argument.

Theorem 7.1 (Finite high tail). *The Hall inequality holds for*

$$2 \cdot 10^9 \leq N < 5 \cdot 10^{11}.$$

Proof. The half-open row endpoints are

$$2,\ 2.3,\ 2.8,\ 4.2,\ 6,\ 12,\ 500 \quad \text{billion}.$$

The endpoints are contiguous, so every N in the displayed range belongs to one half-open interval. On that interval, the finite-prime cutoff fixes F_β, while the nonzero quadratic-residue restrictions and (5) bound Q_β. Every solution is first shown to belong to the corresponding finite list, after which the spacing theorem bounds each root fibre. Lemma 5.5 selects a case, and exact rational comparison puts $D_\beta + F_\beta + Q_\beta$ strictly below $1/25 - 7/(25N)$. Thus (3) follows. □

Theorem 7.2 (Unbounded high tail). *The Hall inequality holds for every*

$$N \geq 5 \cdot 10^{11}.$$

Proof. Use split 120, so large prime squares are those with $p > N/120$ and their complementary quotients satisfy $m \leq 120^2$. At the left endpoint,

$$840^4 \leq 5 \cdot 10^{11} \leq N.$$

Consequently the factor $\left(\sqrt{\sqrt{N}} + 1\right)/N$ in the root count is at most $841/840^4$. An explicit sieve over primes up to 47, together with the quadratic-residue conditions and fibre spacing, gives the common transformed-root bound

$$\frac{8062340}{10^9}.$$

The eight diagonal upper bounds are

$$10^{-9}(26005710, 20944421, 17208292, 16649746, 13003280, 7437057, 6604680, 917036),$$

The diagonal count for each of the eight subsets is periodic after imposing the selected small-prime congruences. Grouping those congruences by their counting moduli gives

$$(1821, 2892, 2627, 2295, 3643, 4590, 3643, 3643).$$

For each diagonal subset, the relevant residue classes are counted modulo its displayed modulus; the associated reciprocal-square tail and endpoint term give the corresponding entry in the diagonal vector. The split at 120, together with the sieve over primes up to 47 and the spacing inequality after (5), gives the common root bound. Substitution in the ten case formulas puts every total below the target at $5 \cdot 10^{11}$. The prime-count, reciprocal-square, fibre-spacing, and endpoint terms are all bounded by functions that do not increase after this endpoint. Therefore the same inequalities apply for every larger N and rule out a strict Hall defect. □

Theorem 7.3 (The range from two billion onward). *The Hall inequality, and hence the exact upper bound, holds for every $N \geq 2 \cdot 10^9$.*

Proof. Theorem 7.1 covers $2 \cdot 10^9 \leq N < 5 \cdot 10^{11}$, and Theorem 7.2 covers $N \geq 5 \cdot 10^{11}$. □

8 Assembly of the exact bound

Theorem 8.1 (The range from forty million onward)**.** *The Hall inequality, and hence the exact upper bound, holds for every* $N \geq 4 \cdot 10^7$.

Proof. Theorems 6.1, 6.2, and 7.3 cover, respectively,

$$[4 \cdot 10^7, 2 \cdot 10^8), \qquad [2 \cdot 10^8, 2 \cdot 10^9), \qquad [2 \cdot 10^9, \infty).$$

□

Theorem 8.2 (The range from five million onward)**.** *The Hall inequality, and hence the exact upper bound, holds for every* $N \geq 5 \cdot 10^6$.

Proof. Theorems 5.3, 5.4, and 5.6 cover

$$[5 \cdot 10^6, 10^7), \quad [10^7, 2 \cdot 10^7), \quad [2 \cdot 10^7, 4 \cdot 10^7),$$

and Theorem 8.1 covers the remaining unbounded interval. These four half-open intervals are pairwise disjoint and cover $[5 \cdot 10^6, \infty)$. □

Proof of Theorem 1.1. Theorem 3.1 gives the upper bound for $N \leq 5 \cdot 10^6$, and Theorem 8.2 gives it for $N \geq 5 \cdot 10^6$. Proposition 1.2 proves sharpness. The Hall formulation is equivalent by Theorem 2.2. □

9 Formal verification

The proof has been formalized in Lean 4 with Mathlib. The formal statement quantifies over every ordered pair $a, b \in A$, including $a = b$, and uses the same mod-25 progressions, Hall completion, residual set, valuation classes, interval endpoints, and rational bounds as the manuscript. The final declaration is

`Erdos848.PaperGeneratedCertificateProvider.all_N`.

Together with the formal sharpness result and the equivalence between the Hall and original formulations, it proves Theorem 1.1. The accompanying theorem map records the formal declarations used in the final interval assembly.

Finite searches are used only to produce explicit witnesses and finite lists. Within the formal proof these objects are ordinary data: general theorems verify prefix coverage, residue-class exhaustion, interval coverage, congruence restrictions, root-fibre bounds, and the exact rational inequalities. The final kernel audit reports only

`[propext, Classical.choice, Quot.sound]`,

the standard classical and quotient dependencies used by Lean and Mathlib.

The version-independent Zenodo concept DOI for the formal archive is `10.5281/zenodo.21750213`. The archive contains the theorem map, the pinned Lean toolchain and Mathlib dependency graph, the checked finite data, and the reproduction instructions.

Disclosure and verification statement

During the preparation of this work, the author used an author-developed local implementation of the Proof Engine framework for AI-assisted mathematical research. The implementation operates through replaceable AI-agent and harness components; the present work used multiple AI tools, including OpenAI Codex, within that implementation. The framework is described in *Proof Engine Infrastructure* [3]. The author reviewed and edited the resulting material and assumes full responsibility for the definitions, statements, proofs, computations, citations, code, and exposition.

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References

[1] Pierre Dusart. Estimates of some functions over primes without R.H., 2010.

[2] Paul Erdős. Some of my favourite problems in various branches of combinatorics. *Le Matematiche*, 47(2):231–240, 1992.

[3] Alex Chengyu Li. Proof engine infrastructure: A fail-closed claim-graph method for accountable AI-assisted mathematical research. SSRN Scholarly Paper 7237460, 2026. Available at https://doi.org/10.2139/ssrn.7237460.

[4] J. Barkley Rosser and Lowell Schoenfeld. Approximate formulas for some functions of prime numbers. *Illinois Journal of Mathematics*, 6:64–94, 1962.

[5] Mehtaab Sawhney. On $A \subseteq [N]$ such that $ab + 1$ is never squarefree for $a, b \in A$. Preprint. Available at https://www.math.columbia.edu/~msawhney/Problem_848.pdf, 2025.

[6] Nat Sothanaphan. An explicit threshold in Erdős problem 848. Preprint dated March 24. Available at https://drive.google.com/file/d/1ujhm4_WYpgRV_rd1rJXIfHyvx16COEKe/view, 2026.